

LLM-Identity Decomposition: Per-Model Asymmetry within HA and AA

Does dyad asymmetry depend on which LLM is in the conversation?

PENPAL Project

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1 Overview

The cross-condition notebooks treat HA and AA as monoliths. This notebook decomposes both by which LLM is in the dyad and asks two questions per metric:

1. **Per-LLM magnitude.** Does unsigned asymmetry A_m vary across the four LLMs within HA? Within AA?
2. **Per-LLM direction.** Does signed slot-asymmetry vary across LLMs (i.e. are some models more likely to drive in a particular direction)?

Sample sizes per LLM are small (HA: ~22-29 stories per model; AA: 20 per model), so reported tests are non-parametric (Kruskal-Wallis + pairwise Wilcoxon BH-adjusted). Bootstrap CIs are also reported.

Where it makes sense, the same-model rows in HA and AA are placed side-by-side: does Claude paired with a human exhibit the same asymmetry profile as Claude paired with itself?

2 Build per-metric per-story tables

```
## Per-story tables built:
##   valence: 140 (HA=100, AA=40)
```

novelty: 140 (HA=100, AA=40)
semexp: 137 (HA=97, AA=40)

3 Within-HA: per-LLM Asymmetry

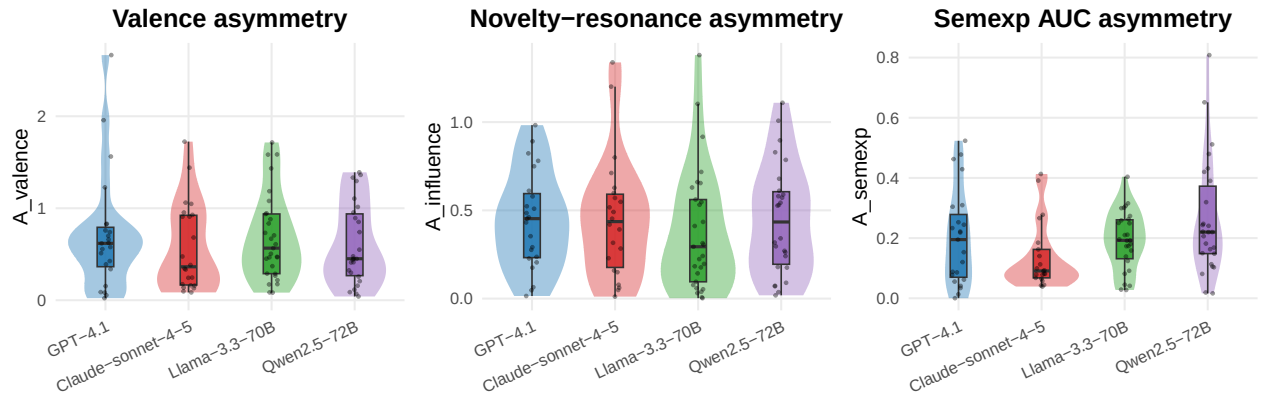
3.1 Magnitude

Table 1: Per-LLM unsigned asymmetry magnitude within HA

metric	llm	n	mean	median	sd
A_valence	GPT-4.1	23	0.712	0.618	0.633
A_valence	Claude-sonnet-4-5	22	0.574	0.364	0.477
A_valence	Llama-3.3-70B	29	0.682	0.566	0.468
A_valence	Qwen2.5-72B	26	0.604	0.451	0.440
A_influence	GPT-4.1	23	0.442	0.453	0.275
A_influence	Claude-sonnet-4-5	22	0.455	0.436	0.348
A_influence	Llama-3.3-70B	29	0.379	0.294	0.346
A_influence	Qwen2.5-72B	26	0.447	0.434	0.312
A_semexp	GPT-4.1	23	0.200	0.195	0.158
A_semexp	Claude-sonnet-4-5	21	0.137	0.091	0.110
A_semexp	Llama-3.3-70B	27	0.192	0.192	0.098
A_semexp	Qwen2.5-72B	26	0.264	0.220	0.192

Table 2: Within-HA Kruskal-Wallis: $A_m \sim \text{llm}$

metric	kw_p
A_valence	0.7988
A_influence	0.6515
A_semexp	0.0521



3.2 Direction

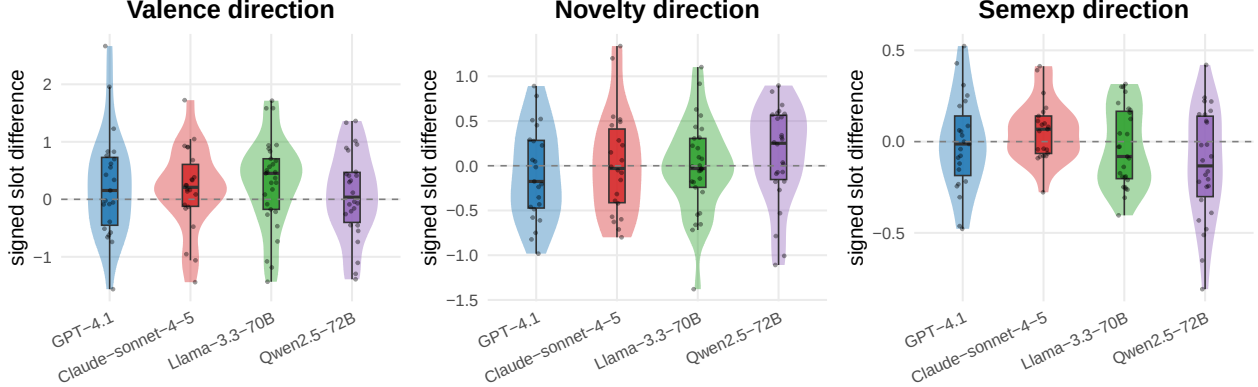
Per-LLM directional stability of the SIGNED slot difference within HA. If the human-side identity is what drives directionality, all four LLMs should look similar (same signed direction). If specific LLMs shape directionality (e.g., Claude is more deferential than Qwen), the stability index will differ across models.

Table 3: Within-HA directional stability of signed asymmetry per LLM

metric	llm	n	mean	sd	stability	sign_consistency	one_sample_t_p
valence	GPT-4.1	23	0.250	0.930	0.268	0.522	0.211
valence	Claude-sonnet-4-5	22	0.181	0.733	0.247	0.682	0.259
valence	Llama-3.3-70B	29	0.293	0.782	0.374	0.690	0.054
valence	Qwen2.5-72B	26	0.080	0.752	0.107	0.500	0.592
novelty	GPT-4.1	23	-0.089	0.521	0.170	0.565	0.422
novelty	Claude-sonnet-4-5	22	0.024	0.581	0.042	0.545	0.845
novelty	Llama-3.3-70B	29	-0.015	0.518	0.030	0.552	0.874
novelty	Qwen2.5-72B	26	0.117	0.540	0.217	0.615	0.278
semexp	GPT-4.1	23	-0.012	0.258	0.045	0.565	0.832
semexp	Claude-sonnet-4-5	21	0.061	0.167	0.363	0.571	0.112
semexp	Llama-3.3-70B	27	-0.035	0.216	0.161	0.593	0.410
semexp	Qwen2.5-72B	26	-0.126	0.304	0.416	0.654	0.044

Table 4: Variance-equality across LLMs (within HA)

metric	brown_forsythe_p
valence	0.6929
novelty	0.8671
semexp	0.0431



4 Within-AA: per-Model Asymmetry

The AA dataset only contains same-model self-pairings (no cross-model pairs). The four levels are therefore pure within-model ai-vs-ai conversations.

4.1 Magnitude

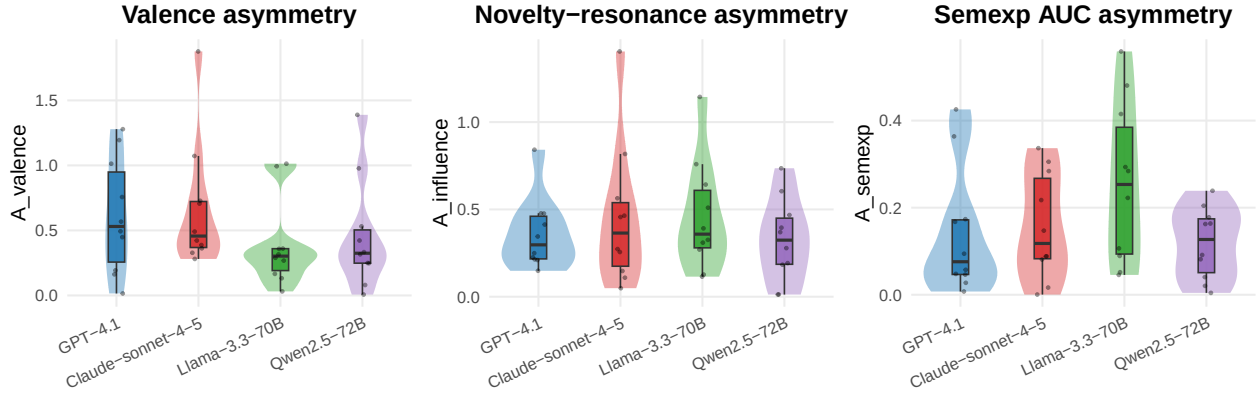
Table 5: Per-model unsigned asymmetry magnitude within AA

metric	llm	n	mean	median	sd
A_valence	GPT-4.1	10	0.612	0.530	0.440
A_valence	Claude-sonnet-4-5	10	0.666	0.457	0.490

metric	llm	n	mean	median	sd
A_valence	Llama-3.3-70B	10	0.393	0.302	0.338
A_valence	Qwen2.5-72B	10	0.455	0.325	0.423
A_influence	GPT-4.1	10	0.360	0.297	0.206
A_influence	Claude-sonnet-4-5	10	0.454	0.365	0.408
A_influence	Llama-3.3-70B	10	0.459	0.358	0.317
A_influence	Qwen2.5-72B	10	0.325	0.324	0.238
A_semexp	GPT-4.1	10	0.141	0.076	0.145
A_semexp	Claude-sonnet-4-5	10	0.157	0.118	0.122
A_semexp	Llama-3.3-70B	10	0.255	0.253	0.185
A_semexp	Qwen2.5-72B	10	0.119	0.127	0.082

Table 6: Within-AA Kruskal-Wallis: $A_m \sim \text{model}$

metric	kw_p
A_valence	0.2257
A_influence	0.8430
A_semexp	0.2783



4.2 Direction

Table 7: Within-AA directional stability of signed asymmetry per model

metric	llm	n	mean	sd	stability	sign_consistency	one_sample_t_p
valence	GPT-4.1	10	0.484	0.591	0.819	0.8	0.029
valence	Claude-sonnet-4-5	10	-0.224	0.822	0.273	0.6	0.411
valence	Llama-3.3-70B	10	-0.065	0.530	0.122	0.7	0.708
valence	Qwen2.5-72B	10	0.150	0.620	0.242	0.5	0.463
novelty	GPT-4.1	10	-0.047	0.429	0.109	0.5	0.737
novelty	Claude-sonnet-4-5	10	-0.204	0.591	0.346	0.6	0.302
novelty	Llama-3.3-70B	10	0.103	0.568	0.182	0.5	0.579
novelty	Qwen2.5-72B	10	0.040	0.415	0.097	0.6	0.766
semexp	GPT-4.1	10	-0.025	0.206	0.122	0.5	0.708
semexp	Claude-sonnet-4-5	10	-0.082	0.186	0.441	0.8	0.196
semexp	Llama-3.3-70B	10	0.063	0.319	0.196	0.6	0.550
semexp	Qwen2.5-72B	10	0.015	0.149	0.100	0.5	0.759

Table 8: Variance-equality across models (within AA)

metric	brown_forsythe_p
valence	0.6494
novelty	0.7260
semexp	0.1801

5 Same Model Across Conditions: HA vs AA per-LLM

For each LLM, place its HA stories (paired with a human) next to its AA stories (paired with itself). Asks whether a given model’s asymmetry profile depends on partner type.

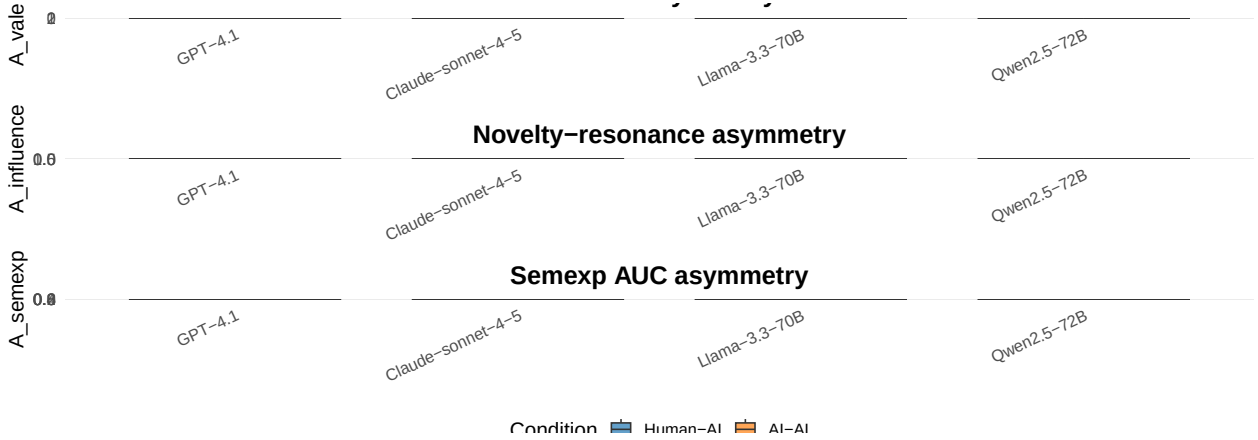


Table 9: HA vs AA per-LLM Wilcoxon (BH-adjusted within metric)

metric	llm	n_HA	n_AA	median_HA	median_AA	cliff_delta	wilcox_p	p_BH
valence	GPT-4.1	23	10	0.618	0.530	0.043	0.862	0.862
valence	Claude-sonnet-4-5	22	10	0.364	0.457	-0.236	0.305	0.450
valence	Llama-3.3-70B	29	10	0.566	0.302	0.393	0.069	0.275
valence	Qwen2.5-72B	26	10	0.451	0.325	0.215	0.337	0.450
novelty	GPT-4.1	23	10	0.453	0.297	0.209	0.363	0.484
novelty	Claude-sonnet-4-5	22	10	0.436	0.365	0.045	0.857	0.857
novelty	Llama-3.3-70B	29	10	0.294	0.358	-0.214	0.332	0.484
novelty	Qwen2.5-72B	26	10	0.434	0.324	0.208	0.355	0.484
semexp	GPT-4.1	23	10	0.195	0.076	0.261	0.253	0.505
semexp	Claude-sonnet-4-5	21	10	0.091	0.118	-0.114	0.633	0.633
semexp	Llama-3.3-70B	27	10	0.192	0.253	-0.193	0.389	0.519
semexp	Qwen2.5-72B	26	10	0.220	0.127	0.515	0.017	0.068

6 Discussion-friendly summary tables

Quick-read summary across all metrics: which LLMs (if any) stand out on magnitude, on direction, and on partner-sensitivity?

Table 10: Directional stability per LLM, by condition (Human-AI vs AI-AI)

condition	metric	llm	n	stability	sign_consistency	one_sample_t_p
AI-AI	novelty	GPT-4.1	10	0.109	0.500	0.737
AI-AI	novelty	Claude-sonnet-4-5	10	0.346	0.600	0.302
AI-AI	novelty	Llama-3.3-70B	10	0.182	0.500	0.579
AI-AI	novelty	Qwen2.5-72B	10	0.097	0.600	0.766
AI-AI	semexp	GPT-4.1	10	0.122	0.500	0.708
AI-AI	semexp	Claude-sonnet-4-5	10	0.441	0.800	0.196
AI-AI	semexp	Llama-3.3-70B	10	0.196	0.600	0.550
AI-AI	semexp	Qwen2.5-72B	10	0.100	0.500	0.759
AI-AI	valence	GPT-4.1	10	0.819	0.800	0.029
AI-AI	valence	Claude-sonnet-4-5	10	0.273	0.600	0.411
AI-AI	valence	Llama-3.3-70B	10	0.122	0.700	0.708
AI-AI	valence	Qwen2.5-72B	10	0.242	0.500	0.463
Human-AI	novelty	GPT-4.1	23	0.170	0.565	0.422
Human-AI	novelty	Claude-sonnet-4-5	22	0.042	0.545	0.845
Human-AI	novelty	Llama-3.3-70B	29	0.030	0.552	0.874
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Human-AI	semexp	Qwen2.5-72B	26	0.416	0.654	0.044
Human-AI	valence	GPT-4.1	23	0.268	0.522	0.211
Human-AI	valence	Claude-sonnet-4-5	22	0.247	0.682	0.259
Human-AI	valence	Llama-3.3-70B	29	0.374	0.690	0.054
Human-AI	valence	Qwen2.5-72B	26	0.107	0.500	0.592

7 Summary

What this notebook adds:

- **Within-HA per-LLM:** tests whether the four LLMs differ in unsigned asymmetry A_m (KW + per-LLM bootstrap-style CIs in the magnitude table) and in directional stability of the signed slot difference. If the HA-specific identity-tied directionality reading from the main notebook holds, all four LLMs should show high stability and a consistent sign on each metric. If it breaks down for one LLM, that LLM contributes less to the HA-distinct signature.
- **Within-AA per-model:** tests whether self-pairings of different LLMs produce different asymmetry profiles. A pure mathematical-baseline reading predicts no difference; a model-specific stylistic reading predicts variation (e.g., Claude self-pair more aligning, Qwen self-pair more divergent).
- **HA vs AA per-LLM:** holds the LLM constant and varies the partner. Same-model HA-vs-AA differences (Wilcoxon + Cliff’s δ , BH-adjusted per metric) isolate the effect of swapping a human partner for a same-LLM partner.

Article-use notes: - Use the Within-HA per-LLM table to support a sentence in the Discussion on whether the HA finding is model-general or model-driven. - Use Within-AA per-LLM to qualify the structural-null result: if AA’s rubber-band excess is concentrated in one model (likely Claude or GPT), this reads as model-specific homogenization, not “AI-AI dyads in general”. - Use HA vs AA per-LLM as a partner-sensitivity panel that complements the three primary cross-condition tests.

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